

24PY901
Quantum Principles, Structures and Computing
Module 1 – Unit 1

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Assignment 2: Physics of Smartphone

Sensors, Communication, Interface, Internals, and Energy

References:

1. Avadhanulu, Chapters 24. Quantum Hypothesis, 26. Matter waves, 27. Quantum mechanics
2. World Wide Web

Instructions:

- Choose the **most appropriate answer** for each question.
- Some technologies may involve more than one physics concept.
- Use this questionnaire to relate physics concepts to real-world devices you use daily.

Sensors

1. A smartphone camera sensor converts incident light into electrical signals. Which physics concept best explains this?
 - (a) Photoelectric effect
 - (b) Compton scattering
 - (c) Diffraction
 - (d) Thermionic emission
2. Touchscreens detect finger position by sensing changes in electric fields. This technology primarily relies on:
 - (a) Capacitance
 - (b) Electromagnetic induction
 - (c) Piezoelectric effect
 - (d) Hall effect
3. Automatic screen rotation in smartphones is achieved using:
 - (a) Accelerometer (Newton's laws)
 - (b) Gyroscope (Angular momentum)
 - (c) Both (a) and (b)
 - (d) Doppler effect
4. Fingerprint sensors in smartphones operate using:
 - (a) Optical reflection
 - (b) Electrostatics
 - (c) Semiconductor physics
 - (d) All of the above
5. Face recognition systems often use infrared depth sensing. The underlying principle involved is:
 - (a) Interference of light
 - (b) Time-of-flight of photons
 - (c) Polarization
 - (d) Photoelectric emission

Communication

6. Mobile phones communicate with base stations using:
 - (a) Radio / microwave electromagnetic waves
 - (b) Infrared radiation
 - (c) Visible light
 - (d) X-rays
7. Mobile signals can bend around buildings due to:
 - (a) Wave nature of electromagnetic radiation
 - (b) Particle nature of photons
 - (c) Relativistic effects
 - (d) Quantum tunneling
8. Noise-cancelling headphones work by producing sound waves of opposite phase. This phenomenon is known as:
 - (a) Constructive interference
 - (b) Destructive interference
 - (c) Diffraction
 - (d) Polarization

Interface

9. Light emission in LED and OLED displays occurs due to:
 - (a) Incandescence
 - (b) Electroluminescence
 - (c) Photoelectric effect
 - (d) Blackbody radiation
10. Change in appearance of phone screens at large viewing angles is mainly due to:
 - (a) Polarization effects
 - (b) Reflection and refraction
 - (c) Dispersion
 - (d) Scattering
11. Blue-light filters in smartphones reduce eye strain by modifying:
 - (a) Wavelength range
 - (b) Speed of light
 - (c) Phase of light
 - (d) Polarization only

Internals

12. Proximity sensors in smartphones switch off the display during calls using:
 - (a) Reflection of infrared radiation
 - (b) Photoelectric effect
 - (c) Thermal radiation
 - (d) Doppler shift
13. The functioning of semiconductor chips in mobile phones is based on:
 - (a) Energy band gap
 - (b) Superconductivity
 - (c) Ferromagnetism
 - (d) Nuclear energy levels
14. Performance degradation of smartphones at high temperatures is due to:
 - (a) Increased lattice vibrations
 - (b) Reduced carrier mobility
 - (c) Increased thermal noise
 - (d) All of the above

Energy

15. Heat generation during fast charging of mobile phones is governed by:
- (a) Joule heating (I^2R loss)
 - (b) Stefan–Boltzmann law
 - (c) Wien’s displacement law
 - (d) Faraday’s law
16. Wireless charging of smartphones works on the principle of:
- (a) Electromagnetic induction
 - (b) Photoelectric effect
 - (c) Electrostatic force
 - (d) Hall effect

Concept Mapping

17. Match the following technologies with the most relevant physics concept:

Technology	Physics Concept
Camera sensor Touchscreen Speaker Microphone GPS Battery	

18. Choose any one feature of a smartphone and explain the physics behind its operation in **3–4 lines**.